

Plugging into the Future: An Exploration of Electricity Consumption Patterns using Tableau

CHAPTER – 01

INTRODUCTION

1.1 – Project Overview

The importance of electricity in our day-to-day lives cannot be understated. Understanding electricity consumption patterns is imperative when managing our energy use in a cost-effective manner and making sustainable development plans. In continuation with this objective, the current project titled "Plugging into the Future: An Exploration of Electricity Consumption Patterns" tries to understand the electricity consumption data via Tableau.

The goal of this project involves the analysis and visualization of the electricity consumption data by state/region, year, month, quarter-wise analysis, and more. This helps us understand various parameters related to how the electricity consumption behaves over time.

Through Tableau, we develop visualizations that enable a deeper understanding of consumption patterns, including

- State-wise consumption of electricity
- Region-wise consumption of electricity
- Trends in consumption of electricity by year
- Consumption of electricity month-wise and quarter-wise
- Top N/Bottom N electricity-consuming states
- Consumption of electricity in metro cities
- Comparison of electricity consumption before & after lockdowns
- Total electricity consumption by regions

Through the power of data visualization, stakeholders are enabled to draw meaningful insights based on filtering and comparisons of the available data sets. In addition to making electricity consumption behavior easier to comprehend, this project also enables us to devise strategies to help with sustainable consumption of electricity.